

PROJECT-SUMMARY

1 AI Standards Knowledge Base - Project Summary

Status: ☐ Core documentation complete and ready for PDF generation
Created: December 4, 2025
Total Content: ~160 KB across 8 markdown files (~39,500 words)

1.1 ☐ What's Included

1.1.1 Core Documentation Files

File	Size	Topics
INDEX.md	9.5 KB	Central navigation hub with cross-links, comparison matrices, quick reference
README.md	8.5 KB	Project overview, structure guide, quickstart, requirements
01-Neural-Networks-Fundamentals.md	45.2 KB	NN theory, training, inference, backpropagation, tokenization (C++ & pseudocode)
02-ONNX.md	25.7 KB	ONNX format specification, protobuf structure, use cases, Python examples
03-OpenAPI-AI-Services.md	26.6 KB	OpenAPI 3.x spec, REST patterns, FastAPI examples, async workflows
04-Model-Cards.md	21.5 KB	Model documentation framework, examples, fairness metrics, tooling
11-Other-AI-Standards-Summary.md	19.6 KB	14 additional standards: PMML, CoreML, TorchScript, DVC, Kubeflow, etc.
CONVERT-TO-PDF.md	4.1 KB	PDF generation instructions (3 methods)
Total: 160.7 KB (~39,500 words)		

1.2 ☐ Quick Start

1.2.1 To Read the Documentation

1. **Browse online:** Open any `.md` file in VS Code
2. **Use INDEX.md:** Central hub with all cross-links
3. **Follow links:** Markdown links work in VS Code preview

1.2.2 To Generate PDFs

See `CONVERT-TO-PDF.md` for three methods: 1. **VS Code Extension** (easiest): Install `yzane.markdown-pdf` 2. **Pandoc CLI** (batch): `pandoc *.md -o pdf/*.pdf` 3. **Online Tools** (no installation): Use `pandoc.org/try` or similar

1.3 ☐ Documentation Hierarchy

```
ai-standards-kb/
├── INDEX.md ..... Main navigation (START HERE)
├── README.md ..... Project overview & setup
├── CONVERT-TO-PDF.md ..... PDF generation guide
├── standards/
│   ├── 01-Neural-Networks-Fundamentals.md
│   ├── 02-ONNX.md
│   ├── 03-OpenAPI-AI-Services.md
│   ├── 04-Model-Cards.md
│   └── 11-Other-AI-Standards-Summary.md
├── annexes/
│   └── (future deep-dives)
└── pdf/
    ├── standards/ (output directory)
    └── annexes/ (output directory)
```

1.4 ☐ Standards Documented (Completed)

1.4.1 Deep Dives (Detailed Coverage)

- ☐ **Neural Networks Fundamentals** - Abstract algorithms, theory, implementations
- ☐ **ONNX** - Model interchange format, protobuf structure, cross-framework compatibility
- ☐ **OpenAPI 3.x** - REST API specification for AI services
- ☐ **Model Cards** - Model documentation and governance framework

1.4.2 Summary Coverage (14 Standards)

- ☐ PMML (classical ML, XML)
- ☐ CoreML (Apple devices, Protobuf)
- ☐ PFA (analytics workflows, JSON)
- ☐ NNEF (Khronos neural formats)
- ☐ TorchScript (PyTorch serialization)
- ☐ DVC (data versioning, Git-based)
- ☐ Kubeflow (Kubernetes ML pipelines)
- ☐ Seldon Core (Kubernetes model serving)
- ☐ BentoML (model packaging framework)
- ☐ Ray Serve (distributed inference)
- ☐ TensorRT (NVIDIA optimization)
- ☐ OpenVINO (Intel optimization)
- ☐ Datasheets (dataset documentation)
- ☐ Apache Arrow (columnar data)

1.5 ☐ Key Features

1.5.1 Code Examples

- ☐ **Pseudocode** algorithms with detailed comments
- ☐ **C++ implementations** (neural networks, matrix ops, tokenizers, loss functions)
- ☐ **Python examples** (PyTorch, TensorFlow, FastAPI)
- ☐ **YAML specs** (OpenAPI, Kubernetes manifests)
- ☐ **JSON schemas** (Model Cards, ONNX structures)

1.5.2 Diagrams & Workflows

- ☐ **Mermaid flowcharts** (training pipeline, inference, API interactions)
- ☐ **Architecture diagrams** (system design patterns)
- ☐ **Sequence diagrams** (protocol flows)

1.5.3 Authoritative References

- ☐ Links to official specifications
- ☐ Tool ecosystem with URLs
- ☐ Learning resources and tutorials

- ☐ GitHub repositories and examples

1.5.4 Use Cases & Procedures

- ☐ Procedural workflows with step-by-step instructions
- ☐ Real-world examples (image classification, text generation, batch processing)
- ☐ Deployment patterns and best practices

1.6 ☐ Future Expansions (Planned)

10 additional documentation files are planned for future releases:

Model Serving Standards: - TensorFlow Serving (SavedModel protocol, gRPC/REST) - MLflow (tracking, model registry, project format) - Hugging Face Hub (model cards, safetensors, configs) - gRPC (Protocol Buffers for ML) - KServe (Kubernetes inference service) - Triton Inference Server (model repository, backends)

Embeddings & Retrieval: - Vector Embeddings Standards (formats, dimensions) - Vector Databases (Pinecone, Weaviate, Milvus, etc.) - RAG Patterns (retrieval-augmented generation)

LLM-Specific: - Prompt Engineering Standards (templates, few-shot) - LLM API Standards (OpenAI, Anthropic, schemas)

1.7 ☐ File Organization

```
c:\Users\plachy\Documents\Dev\AI_KB\ai-standards-kb\
├─ Root files (INDEX, README, CONVERT-TO-PDF)
├─ standards/
│   ├── 01-Neural-Networks-Fundamentals.md
│   ├── 02-ONNX.md
│   ├── 03-OpenAPI-AI-Services.md
│   ├── 04-Model-Cards.md
│   └── 11-Other-AI-Standards-Summary.md
├─ annexes/ (reserved for future)
├─ pdf/ (output location for PDFs)
│   ├── standards/ (generated PDFs)
│   └── annexes/ (reserved for future)
└─ generate-pdfs.ps1 (helper script)
```

1.8 ☐ Content Statistics

Metric	Count
Total markdown files	8
Total lines of documentation	~1,300
Total words	~39,500
Code examples	50+
Diagrams (Mermaid)	8+
Cross-links	30+
Authoritative references	40+
Standards documented	18 (4 deep + 14 summary)

1.9 ☐ Completion Status

Task	Status
Directory structure	☐ Complete
Core documentation (4 files)	☐ Complete
Supporting docs (INDEX, README)	☐ Complete
PDF generation helper	☐ Complete
PDF conversion guide	☐ Complete

Task	Status
Phase 1 Total	☐ 100%
Remaining documentation (10 files)	☐ Planned
PDF generation execution	☐ Ready
Extended standards (future)	☐ Planned

1.10 ☐ How to Use This Knowledge Base

1.10.1 For Learning

1. Start with **INDEX.md** for overview
2. Read **README.md** for structure
3. Choose a standard from the index
4. Follow the deep-dive documentation with code examples

1.10.2 For Reference

- Use INDEX.md as a navigation hub
- Each file has a table of contents
- Cross-links enable jumping between related topics
- Code examples are copy-paste ready

1.10.3 For Implementation

- Find procedural use cases in each standard's documentation
- Use provided code examples as starting templates
- Follow tool ecosystem recommendations
- Reference authoritative sources for edge cases

1.10.4 For PDF Reading

1. Convert using one of 3 methods in CONVERT-TO-PDF.md
 2. PDFs include table of contents for easy navigation
 3. External links remain functional in PDFs
 4. Code examples are syntax-highlighted
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1.11 ☐ External Resources

1.11.1 Official Standards

- ONNX Runtime: <https://onnxruntime.ai/>
- OpenAPI Initiative: <https://www.openapis.org/>
- TensorFlow: <https://www.tensorflow.org/>
- PyTorch: <https://pytorch.org/>

1.11.2 Tools & Platforms

- VS Code: <https://code.visualstudio.com/>
 - Pandoc: <https://pandoc.org/>
 - Hugging Face: <https://huggingface.co/>
 - Kubernetes: <https://kubernetes.io/>
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1.12 ☐ Notes

- All code examples are production-ready patterns (not toy examples)
 - Documentation includes both high-level concepts and implementation details
 - Cross-links work in VS Code markdown preview
 - Mermaid diagrams render in most markdown viewers
 - Ready for PDF generation via Pandoc or VS Code extension
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Last Updated: December 4, 2025

Created By: AI Standards Knowledge Base Project

Purpose: Comprehensive reference for AI/ML protocols and integration standards